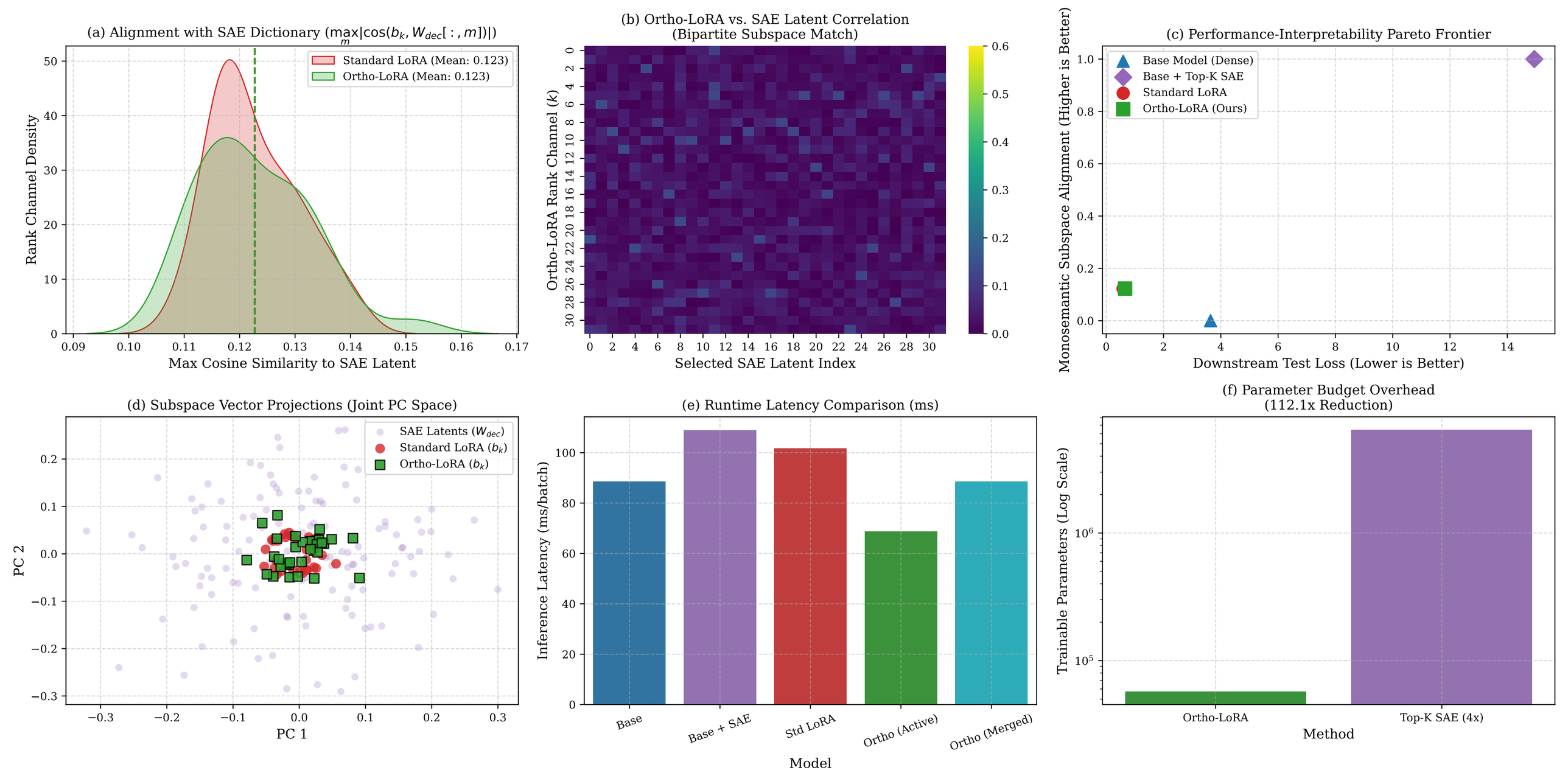

(a) Alignment with SAE Dictionary ($\max_m|\cos(b_k, W_{dec}[:, m])|$)
Standard LoRA (Mean: 0.123)
Ortho-LoRA (Mean: 0.123)
Rank Channel Density
Max Cosine Similarity to SAE Latent
(b) Ortho-LoRA vs. SAE Latent Correlation
(Bipartite Subspace Match)
Ortho-LoRA Rank Channel (k)
Selected SAE Latent Index
(c) Performance-Interpretability Pareto Frontier
Base Model (Dense)
Base + Top-K SAE
Standard LoRA
Ortho-LoRA (Ours)
Monosemantic Subspace Alignment (Higher is Better)
Downstream Test Loss (Lower is Better)
(d) Subspace Vector Projections (Joint PC Space)
SAE Latents (W_{dec})
Standard LoRA (b_k)
Ortho-LoRA (b_k)
PC 1
PC 2
(e) Runtime Latency Comparison (ms)
Inference Latency (ms/batch)
Base
Base + SAE
Std LoRA
Ortho (Active)
Ortho (Merged)
Model
(f) Parameter Budget Overhead
(112.1x Reduction)
Trainable Parameters (Log Scale)
Ortho-LoRA
Top-K SAE (4x)
Method